# Arunabha Majumder

**Mechatronics Engineer | Control Systems, Motion, System Modelling & Integration (MATLAB/Simulink)**

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## SUMMARY

Mechatronics engineer focused on control systems. Designs control strategies, models system dynamics and kinematics, and integrates sensors and actuators into prototypes tested at system level. Achieved 0.3–0.78% tracking error on a non-linear pneumatic actuator and a 25% energy reduction through a new actuation architecture. MATLAB/Simulink, Python.

## KEY RESULTS

- **0.3–0.78%** tracking error with a neural-network gain-scheduled PID controller
- **>25%** energy reduction from a new actuation system concept (Best Paper Award, 2026)
- **<10%** RMS error from a kinematic (Jacobian) force-estimation model

## EXPERIENCE

**Research Engineer – Mechatronic Systems & Control (PhD Project: VIEXO)** | 2023 – Present
*Aalborg University, Aalborg, Denmark*

- Reduced user muscle effort by more than 15%, measured with EMG, by developing and implementing a control strategy (Assist-as-Needed hybrid control) on a mechatronic system, then testing and tuning it at system level.
- Cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by translating functional requirements into a system concept: a hybrid actuation architecture with a parallel spring and a 6 Nm motor.
- Achieved below 10% RMS error in end-effector force estimation, validated against a force sensor, by modelling the kinematics of a 5-bar planar parallel robot (Jacobian) and its variable-stiffness actuators.
- Selected and integrated sensors and actuators (motor, encoder, force sensors, DAQ, embedded control) into prototypes designed in SolidWorks and built through 3D printing.
- Solved system and integration problems by analysing measurement data from custom test rigs, working closely with electronics, control and biomechanics specialists; mentored Bachelor's students (2023 – 2025).

**MTech Researcher – Control of Non-Linear Pneumatic Actuators (Master's Thesis)** | 2020 – 2022
*CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Reduced position-tracking error to 0.3–0.78%, outperforming classical PID under varying load, by designing a neural-network-based gain-scheduled PID controller in MATLAB/Simulink for a highly non-linear actuator.
- Built an empirical dynamic model from measurement data by fabricating pneumatic muscles to an existing lab design and testing them on the lab's setup (valves, pressure sensors, DAQ) and characterising their load–displacement behaviour.

**Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton** | 2018 – 2019
*Siddaganga Institute of Technology, Tumakuru, India*

- Led CAD design, machining, hands-on assembly and user testing; Best Major Project Award and INR 100,000 TATA Technologies Innovation Grant.

## SKILLS

- **Control engineering:** PID, gain scheduling, neural-network-based control, Assist-as-Needed control, motion control

- **Modelling & analysis:** System dynamics, kinematics, Jacobian-based estimation, empirical modelling from test data
- **Tools:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Sensors & actuators:** Motors, encoders, force and pressure sensors, pneumatic valves, DAQ, embedded control
- **Prototyping:** SolidWorks (CSWA), 3D printing, test rigs
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

## EDUCATION

**Ph.D. Candidate, Mechatronics & Robotics** | 2023 – Present
*Aalborg University, Department of Materials and Production, Aalborg, Denmark*

**M.Tech, Mechatronics** | 2020 – 2022
*AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

**B.Tech, Mechanical Engineering** | 2015 – 2019
*Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India*

## AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.

## REFERENCES

Available upon request.